IPCON® PPA CF

Carbon Fiber Reinforced Polyphthalamide Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.05 mm
Net Filament Weight	0.75 kg ± 0.5%
Filament Length	250 m ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C) or high-temp PC (max heat resistance: 140 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 110 mm
Recommendation for body protection during use	This material contains carbon fiber, which may cause skin irritation. So it is recommended to wear gloves throughout the operation.

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.26 g/cm³
Shrinkage Rate*	Specially-defined	0.15% - 0.44%
Surface Hardness	Shore hardness	88D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	1.30%
Melt Index	290 °C, 2.16 kg	12.4 g/10 min
Melting Temperature	DSC, 10 °C/min	261 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	233 °C
Heat Deflection Temperature	ISO 75, 0.45 MPa	227 °C
Relative Thermal Index (RTI) Max Continuous Service Temperature	Actual working conditions simulation test	145 - 160 °C (dry state) (Related to load, size, shape, wall thickness, rate of water content, etc., and the value after annealing is higher than that before annealing in general)

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	169 MPa
Tensile Strength Z	ISO 527, GB/T 1040	58 MPa
Young's Modulus XY	ISO 527, GB/T 1040	12673 MPa
Young's Modulus Z	ISO 527, GB/T 1040	4269 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	2.2%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	1.2%
Flexural Strength XY	ISO 178, GB/T 9341	256 MPa

Flexural Strength Z	ISO 178, GB/T 9341	80 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	11384 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	3278 MPa
Impact Strength XY	ISO 179, GB/T 1043	34.2 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	4.6 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 290 °C, bed temperature: 100 °C, printing speed: 100 mm/s) and IPCON PPA CF filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. Extreme temperature and humidity resistance (short-term peak of temperature resistance reaches 220 °C); lightweight; super strong and stiff; creep-resistant, ultra low shrinkage and warping, high precision; anti-static (XY direction).
2. ESD parts in the semiconductor and oil and gas industry; substitutes of metal parts in the drone, automobile and machinery industries.

• Recommended Printing Settings

Drying Settings Before Printing	100 - 140 °C, 10 - 14 h (Blast drying oven)
Printing Temperature and Humidity	≤ 120 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for PA
Compatible Printer Type	Enclosed-frame (recommended), open-frame,
AMS or CFS	Not Compatible
Compatible Nozzle Material	Hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.6 mm (recommended) / 0.4 mm / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Apply liquid glue or PVP solid glue stick if the first layer bonding is not firm enough
Nozzle Temperature	280 - 310 °C
Bed Temperature	100 - 120 °C
Fan Speed	0% - 40%
Max Print Speed	250 mm/s

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